

Reference book - Kogge-LoFlamme - Mosca
An Introduction to Quantum Computing

Theorem: $|\psi\rangle \in H_A \otimes H_B$, then there exists a $\{ \varphi_i^A \}$ ONB basis of H_A and $\{ \varphi_i^B \}$ ONB basis of H_B and $\{ p_i \}$ real non negative such that

$$|\psi\rangle = \sum_{i=0}^{mn(n,m)} \sqrt{p_i} |\varphi_i^A\rangle |\varphi_i^B\rangle$$

ex. $|\psi\rangle = \frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle + \frac{1}{2} |11\rangle$

$$\{ |\varphi_0^{(A)}\rangle = |+\rangle, |\varphi_1^{(A)}\rangle = |-\rangle \}$$

$$\{ |\varphi_0^{(B)}\rangle = |+\rangle, |\varphi_1^{(B)}\rangle = |-\rangle \}$$

one Schmidt basis

$$|\psi\rangle = 1 |++\rangle + 0 |--\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)$$

Comments of Dirac notation

$$|i\rangle \otimes |j\rangle = |i\rangle |j\rangle = |ij\rangle$$

$$\underbrace{|x\rangle \otimes |x\rangle \otimes \dots \otimes |x\rangle}_n = |x\rangle^{\otimes n}$$

$$|i\rangle_1 |j\rangle_2$$

$$(|i\rangle_1 |j\rangle_2)^\dagger = \langle i|_1 \langle j|_2$$

the conjugate transp.
doesn't change the order

But physicist often do

$$(|i\rangle_1 |j\rangle_2)^\dagger = \langle j|_2 \langle i|_1$$

$$(|i\rangle_1 |j\rangle_2)^\dagger |k\rangle_1 |l\rangle_2 = \langle i|_1 \langle j|_2 \langle i|_1 k\rangle_1 |l\rangle_2 = \langle i|k\rangle \langle j|l\rangle$$

For the outer product

$$(|i\rangle_1 \otimes |j\rangle_2) (\langle k|_1 \otimes \langle l|_2) = |i\rangle_1 \langle k|_1 \otimes |j\rangle_2 \langle l|_2$$

Composite systems

$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\underline{|11\rangle} = |1\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$|n\rangle = \underbrace{\hspace{1cm}}_{\log n} \rangle$$

$$|7\rangle = |111\rangle = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \quad 8$$

A 2-qubit state cannot ~~be~~ always be written as tensor product of two 1-qubit system.

$$|0\rangle \rightarrow \boxed{X} \rightarrow |1\rangle$$

$$|1\rangle \rightarrow \boxed{H} \rightarrow |+\rangle$$

Bell-state (EPR state)
Einstein Podolsky Rosen

$$|b_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad \text{entangled state}$$

$$|\varphi_1\rangle = \alpha|0\rangle + \beta|1\rangle \quad |\varphi_2\rangle = \gamma|0\rangle + \delta|1\rangle$$

We cannot find any $\alpha, \beta, \gamma, \delta$ such that

$$|\varphi_1\rangle \otimes |\varphi_2\rangle = |b_{00}\rangle$$

$$\alpha\gamma|00\rangle + \underline{\beta\gamma|10\rangle} + \underline{\alpha\delta|01\rangle} + \beta\delta|11\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

$$\begin{cases} \alpha\gamma = \frac{1}{\sqrt{2}} \\ \beta\gamma = 0 \\ \alpha\delta = 0 \\ \beta\delta = \frac{1}{\sqrt{2}} \end{cases}$$

We can derive the existence of entangled states also from a counting argument.

1 qubit

$$\alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}$$

$$|\alpha|^2 + |\beta|^2 = 1$$

$$4 - 2 = 2$$

↑
normalization,
global phase

2 qubit

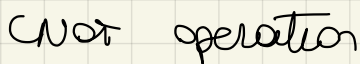
$$\dim(H_2 \otimes H_2)$$

$$2 \cdot 4 - 2 = 6$$

n qubit

$$2 \cdot 2^n - 2 = 2^{n+1} - 2$$

Copying information



Proof: Assume by contradiction that we have U such that

$$U(|x\rangle \otimes |a\rangle) = |x\rangle \otimes |x\rangle$$

cancel - present

$$U^+ U \subseteq I$$

$$\langle x | \langle a | \underbrace{\bigcup_{I=1}^{\infty}}^{\dagger} \bigcup | y \rangle | a \rangle = \langle x | y \rangle \underbrace{\langle a | a \rangle}_{=1} = \langle x | y \rangle$$

$$\langle x | \otimes \langle x | y \rangle \otimes | y \rangle = \langle x | y \rangle \langle x | y \rangle = (\langle x | y \rangle)^2$$

$$\langle x|y \rangle = (\langle x|y \rangle)^2 \begin{cases} \langle x|y \rangle = 0 \\ \langle x|y \rangle = 1 \end{cases}$$

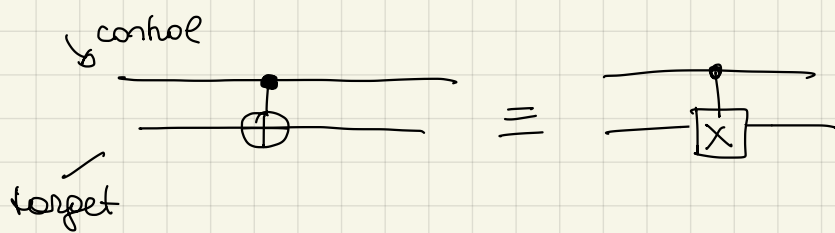
control target

$CNOT(x, y) = \begin{cases} x, y \oplus x = (x, y) & x = 0 \\ x, \text{NOT}(y) & x = 1 \end{cases}$

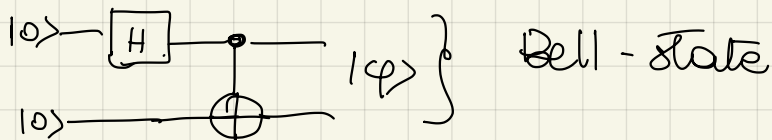
00 01 10 11

00 01 10 11

00	1	0	0	0
01	0	1	0	0
10	0	0	0	1
11	0	0	1	0



$$\begin{aligned} \text{CNOT} \left((\alpha|0\rangle + \beta|1\rangle) |0\rangle \right) &= \text{CNOT} (\alpha|00\rangle + \beta|10\rangle) = \\ &= \alpha \text{CNOT}(|00\rangle) + \beta \text{CNOT}(|10\rangle) = \alpha|00\rangle + \beta|11\rangle \end{aligned}$$

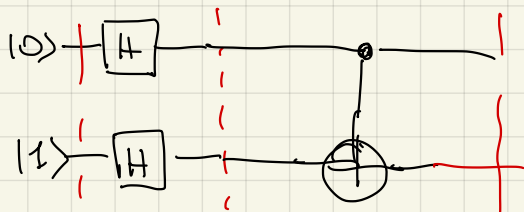


$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 0 & 1 \\ & & 1 & 0 \end{bmatrix} = \begin{bmatrix} I & \\ & X \end{bmatrix}$$

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 0 & 1 \\ & & 1 & 0 \end{bmatrix} \left(\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

$$\downarrow$$

$$\begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$



$\varphi_0 = |0\rangle \otimes |+\rangle$ $\varphi_1 = |+\rangle \otimes |-\rangle$ φ_F

$$\varphi_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$$

$\begin{matrix} |+\rangle & & |-\rangle \end{matrix}$

$$\frac{1}{2} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & 0 & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle - \frac{1}{2} |10\rangle + \frac{1}{2} |11\rangle$$

$$= |--\rangle$$

$$= \phi \frac{1}{2} (|0\rangle - |1\rangle) (|0\rangle - |1\rangle) =$$

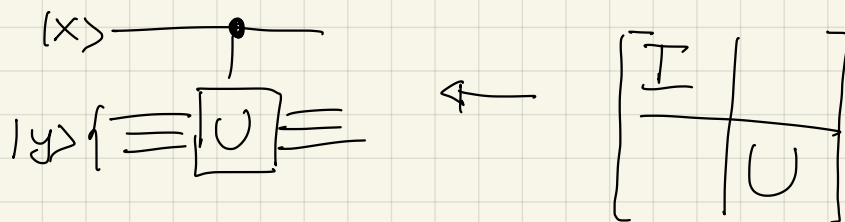
$$= \frac{1}{2} (|00\rangle - |10\rangle - |01\rangle + |11\rangle)$$

the effect of the operation is to change the control qubit and not the target is called phase kickback

$$\text{CNOT}(|x\rangle) = (-1)^x |x\rangle |-\rangle \quad x \in \{0, 1\}$$

$$\text{CNOT}((\alpha|0\rangle + \beta|1\rangle)|-\rangle) = (\alpha|0\rangle - \beta|1\rangle)|-\rangle$$

In general



$$|x\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|x\rangle |y\rangle \xrightarrow{\text{C-U}}$$

$$\text{C-U}(\alpha|0\rangle + \beta|1\rangle |y\rangle) = \alpha|0\rangle |y\rangle + \beta|1\rangle U|y\rangle$$

U_f : 2 qubit unitary operator $f: \{0, 1\} \rightarrow \{0, 1\}$ boolean function

$$U_f |x y\rangle = |x\rangle |y \oplus f(x)\rangle$$

$$U_f |x -\rangle = U_f \left(|x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = U_f \left(\frac{|x0\rangle - |x1\rangle}{\sqrt{2}} \right)$$

$$= \frac{1}{\sqrt{2}} U_f |x0\rangle - \frac{1}{\sqrt{2}} U_f |x1\rangle$$

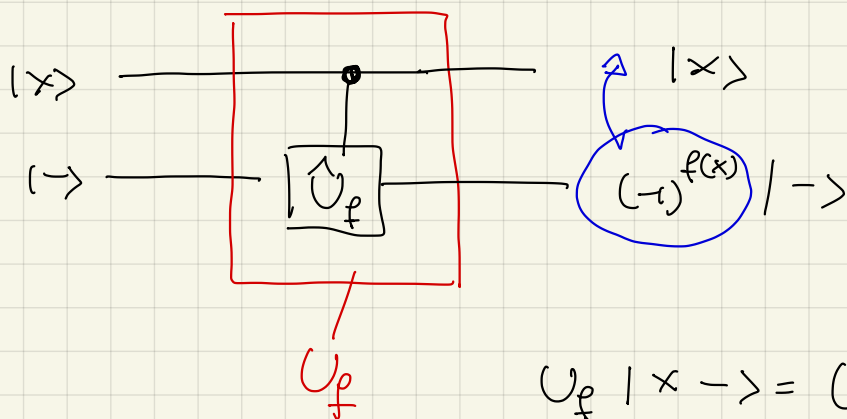
$$= \frac{1}{\sqrt{2}} |x|0 \oplus f(x)\rangle - \frac{1}{\sqrt{2}} |x|1 \oplus f(x)\rangle =$$

$$= |x\rangle \left(\frac{|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}} \right) =$$

$$-f(x)=0 \Rightarrow |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$-f(x)=1 \Rightarrow |x\rangle \left(\frac{|1\rangle - |0\rangle}{\sqrt{2}} \right) = -|x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$\Delta = (-1)^{f(x)} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

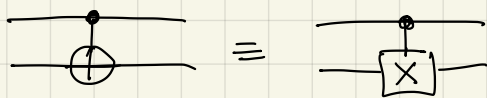


$$U_f |x -\rangle = (-1)^{f(x)} |x -\rangle$$

$|-\rangle$ is an eigenvector corresponding to U_f and the eigenvalue $(-1)^{f(x)}$ kicks back in front of the control

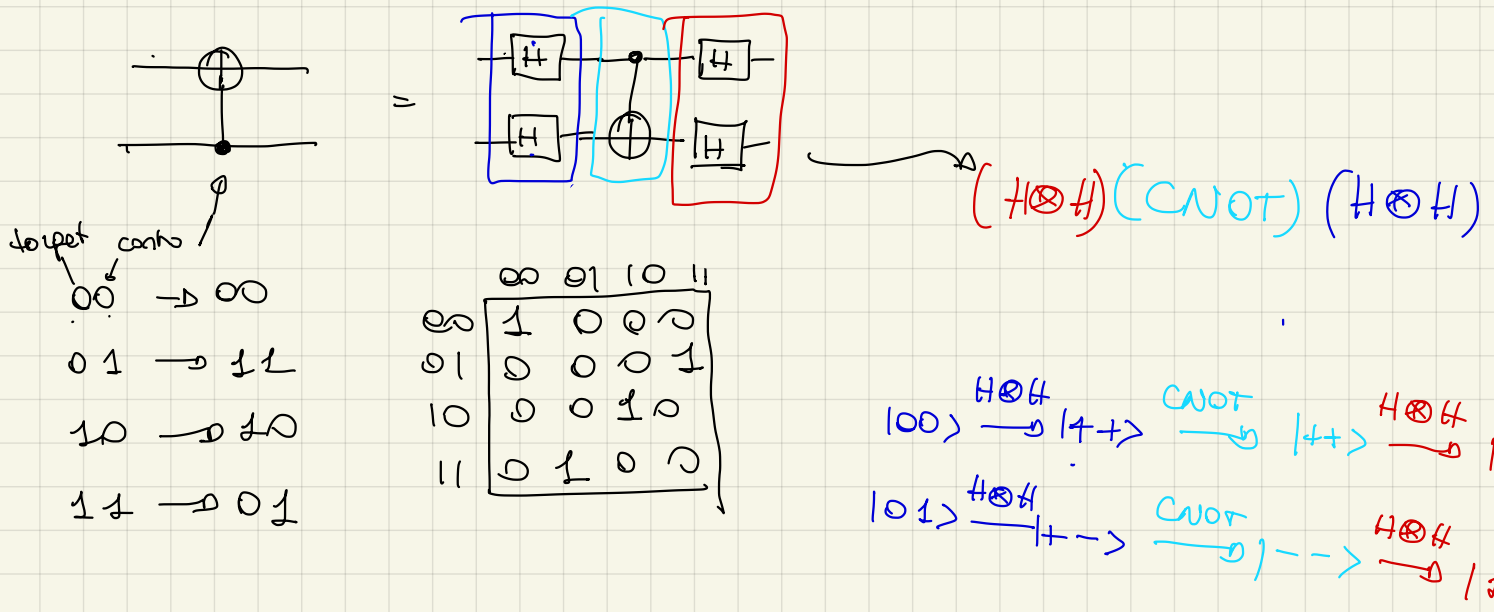
qubit.

Circuits

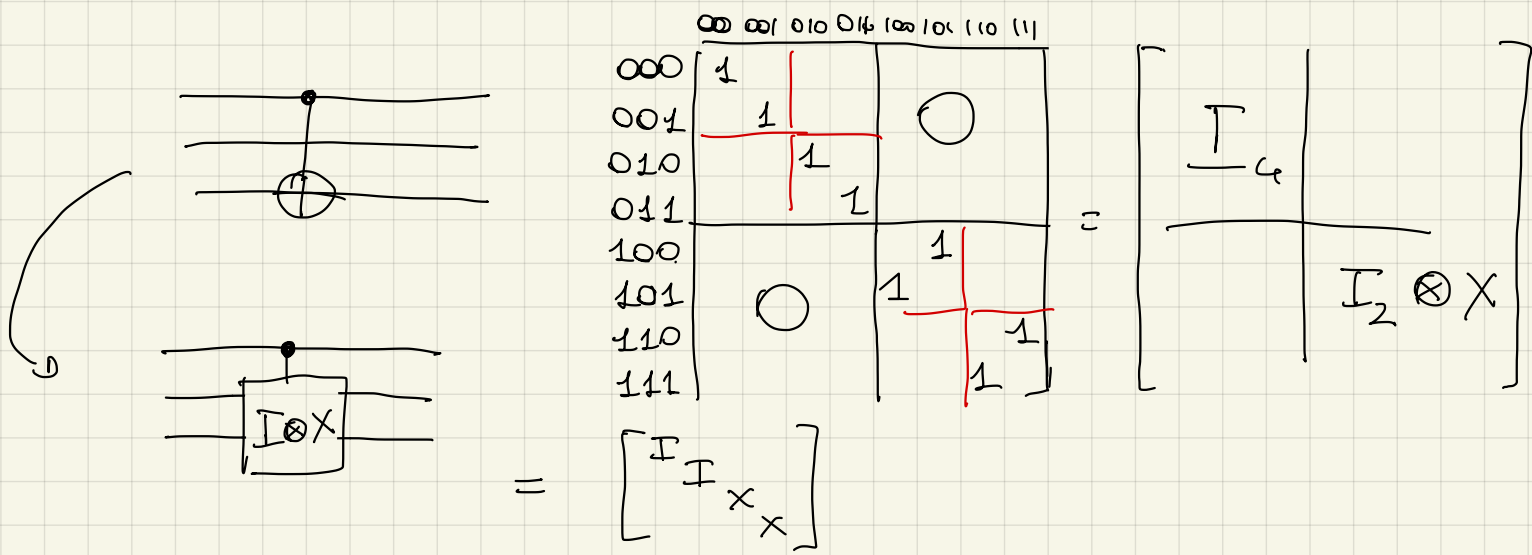


$$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

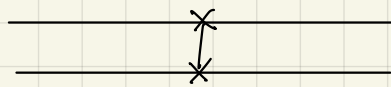
$$I \otimes H = \begin{bmatrix} H & 0 \\ 0 & H \end{bmatrix}$$



More examples

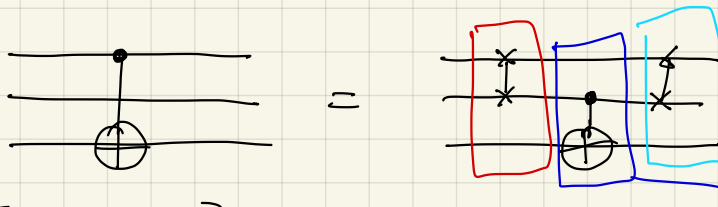


SWAP - GATE

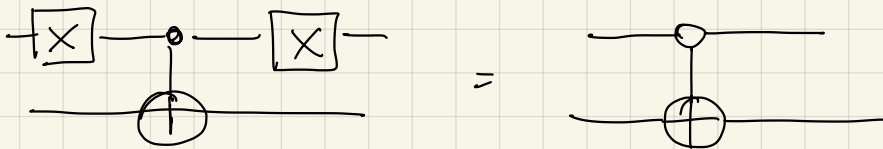


$$\begin{aligned} |00\rangle &\rightarrow |00\rangle \\ |01\rangle &\rightarrow |10\rangle \\ |10\rangle &\rightarrow |01\rangle \\ |11\rangle &\rightarrow |11\rangle \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} I & & & \\ & I & & \\ & & X & \\ & & & X \end{bmatrix} = (\text{SWAP} \otimes I) (I \otimes \text{CNOT}) (\text{SWAP} \otimes I)$$



if control = 0
apply not on target

$$\begin{pmatrix} 1 & & & \\ & 0 & 1 & \\ & 1 & 0 & \\ & & & 1 \end{pmatrix} \begin{pmatrix} \alpha & \gamma \\ \alpha & \delta \\ \beta & \gamma \\ \beta & \delta \end{pmatrix} = \begin{pmatrix} \alpha & \gamma \\ \beta & \gamma \\ \alpha & \delta \\ \beta & \delta \end{pmatrix} \therefore \begin{bmatrix} \gamma \\ \delta \end{bmatrix} \oplus \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \oplus \begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha & \gamma \\ \alpha & \delta \\ \beta & \gamma \\ \beta & \delta \end{bmatrix}$$